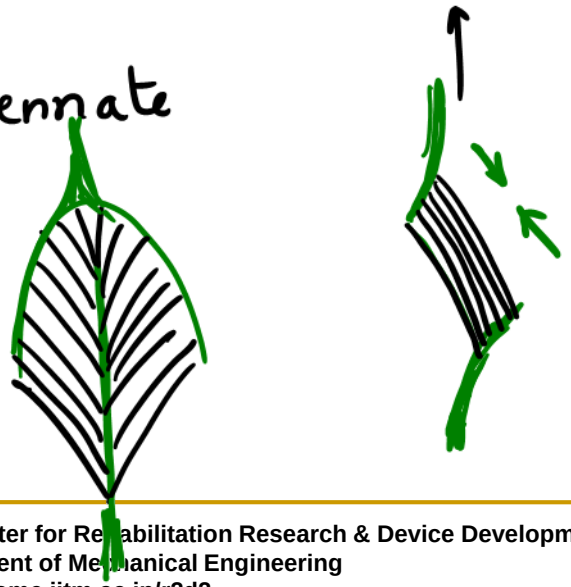
Skeletal muscles - structure

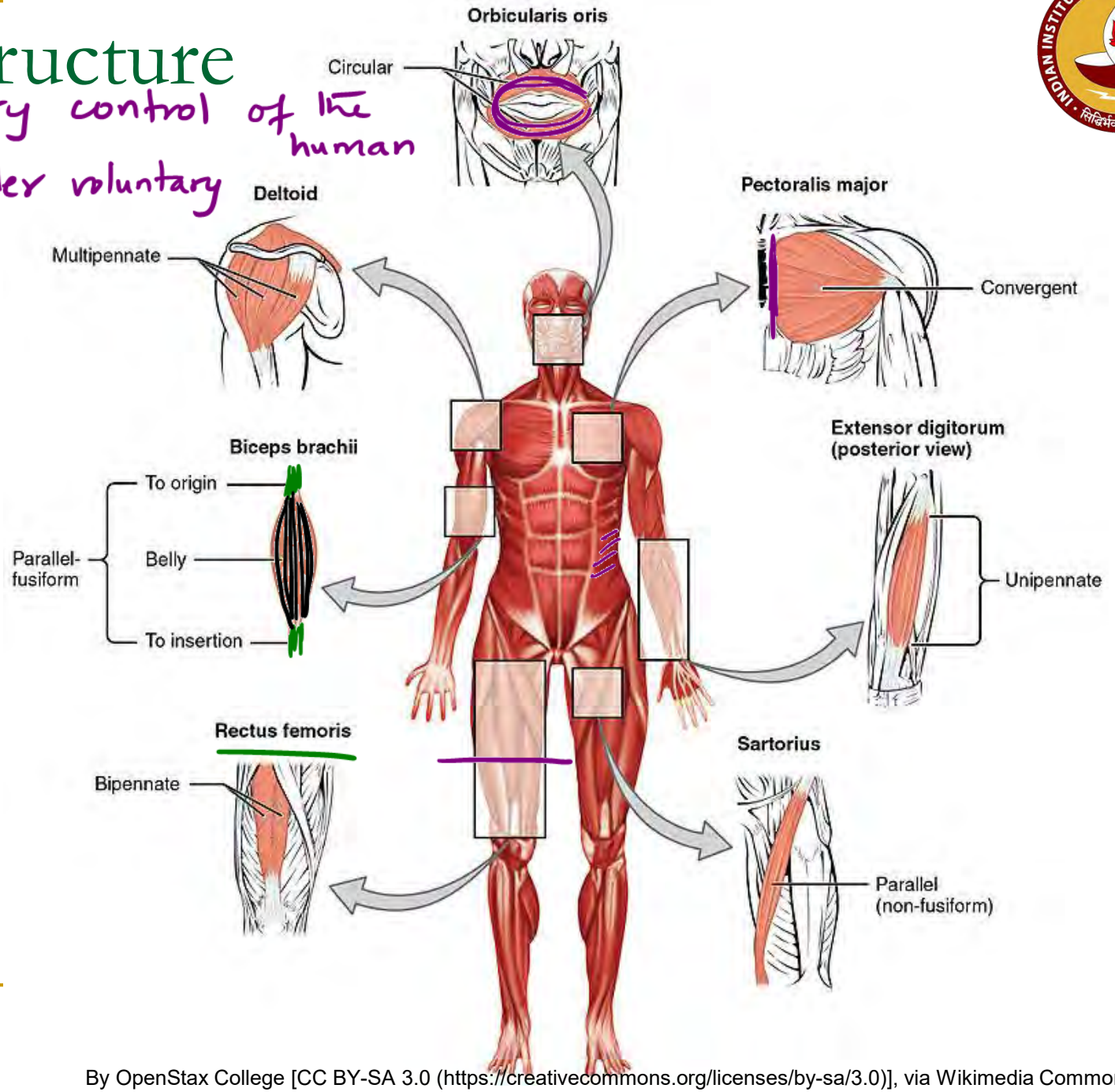
Striated — Skeletal — voluntary control of the human
Cardiac
Smooth — not under voluntary control

- Groups contained in compartments defined by fibrous tissue called fascia
- Fiber arrangements
  - Parallel
  - Pennate or Penniform

Fascicles

Unipennate





TTK Center for Rehabilitation Research & Device Development (R2D2)
Department of Mechanical Engineering
https://home.iitm.ac.in/r2d2

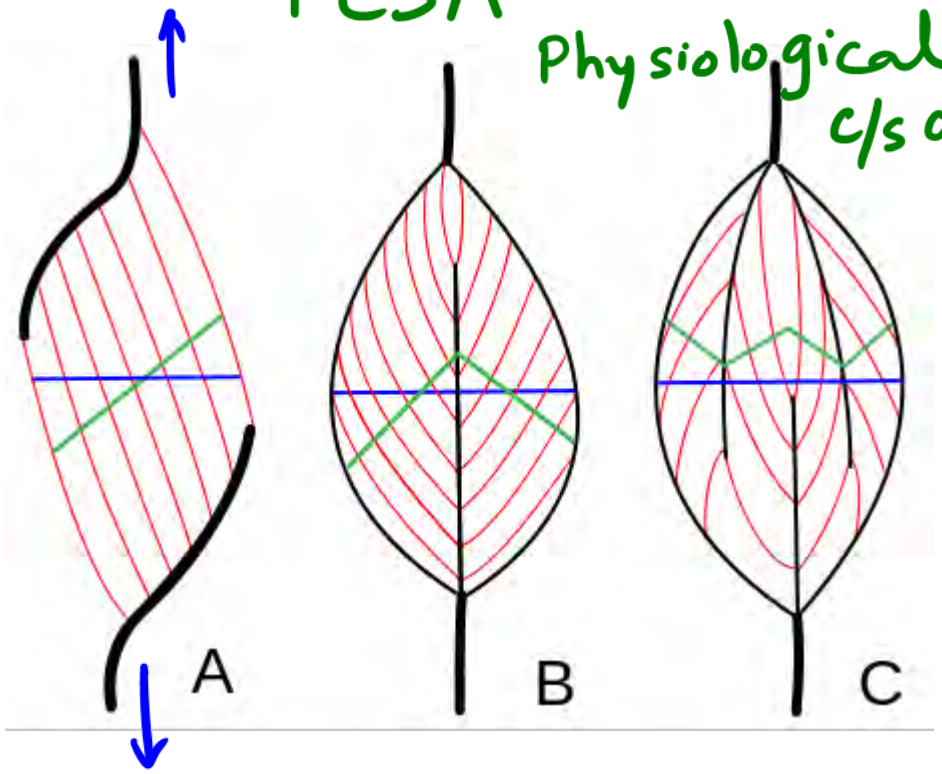
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Skeletal muscles

Anatomical c/s area

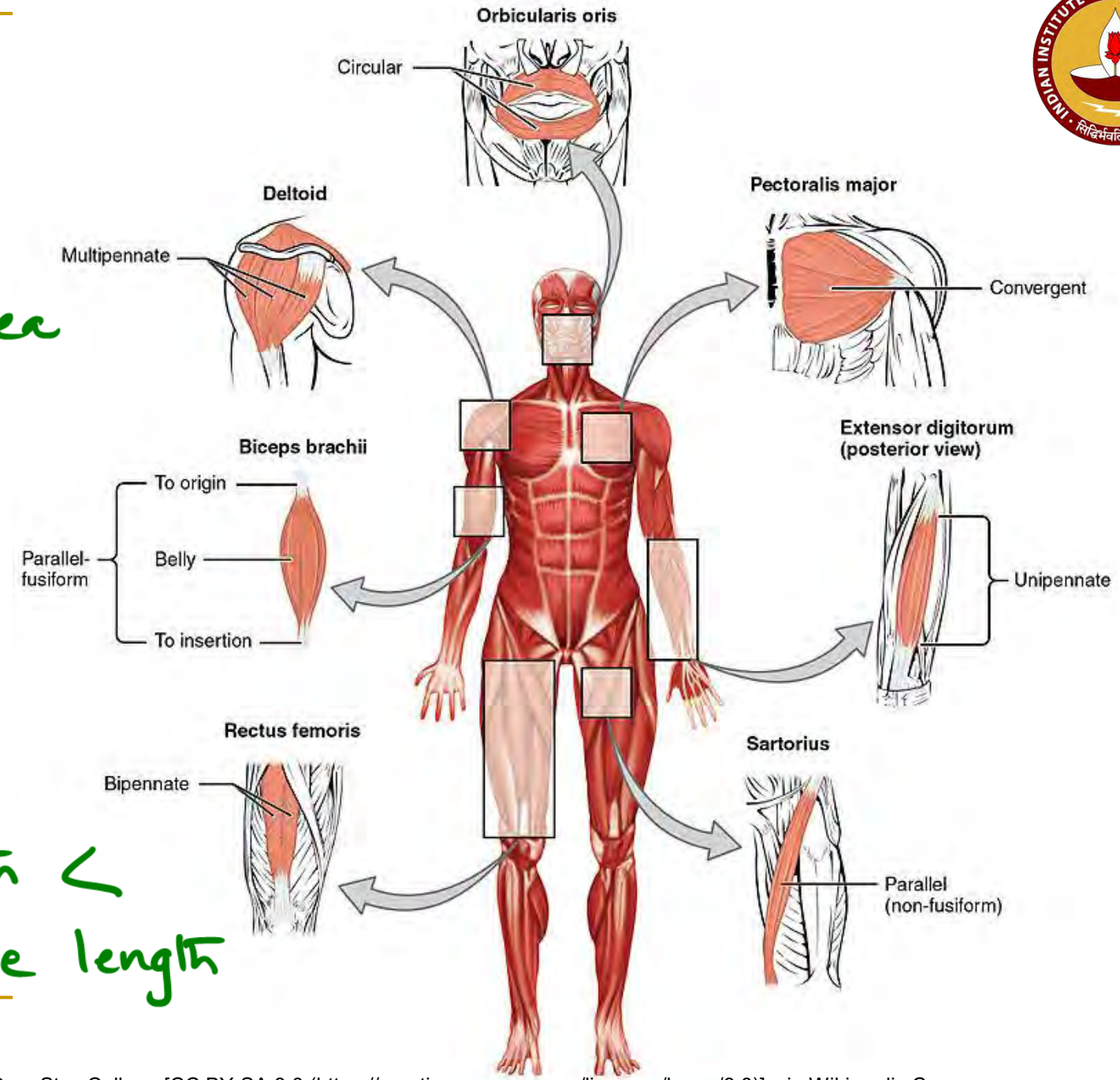
PCSA

Physiological c/s area



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Fiber length < muscle length

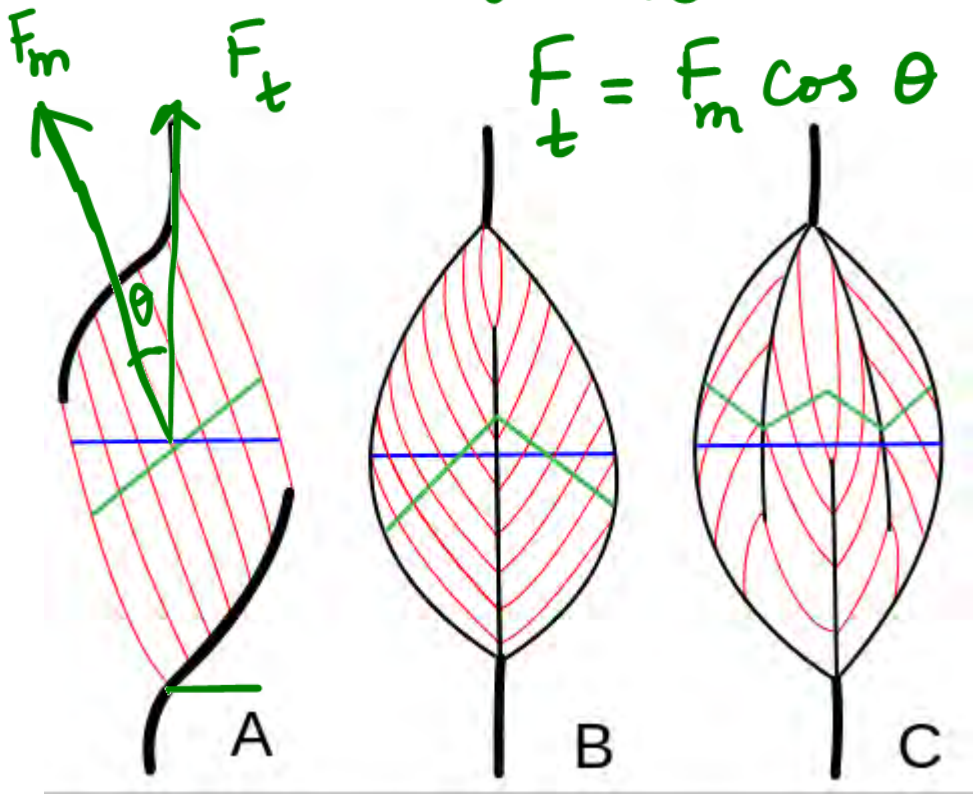


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Skeletal muscles

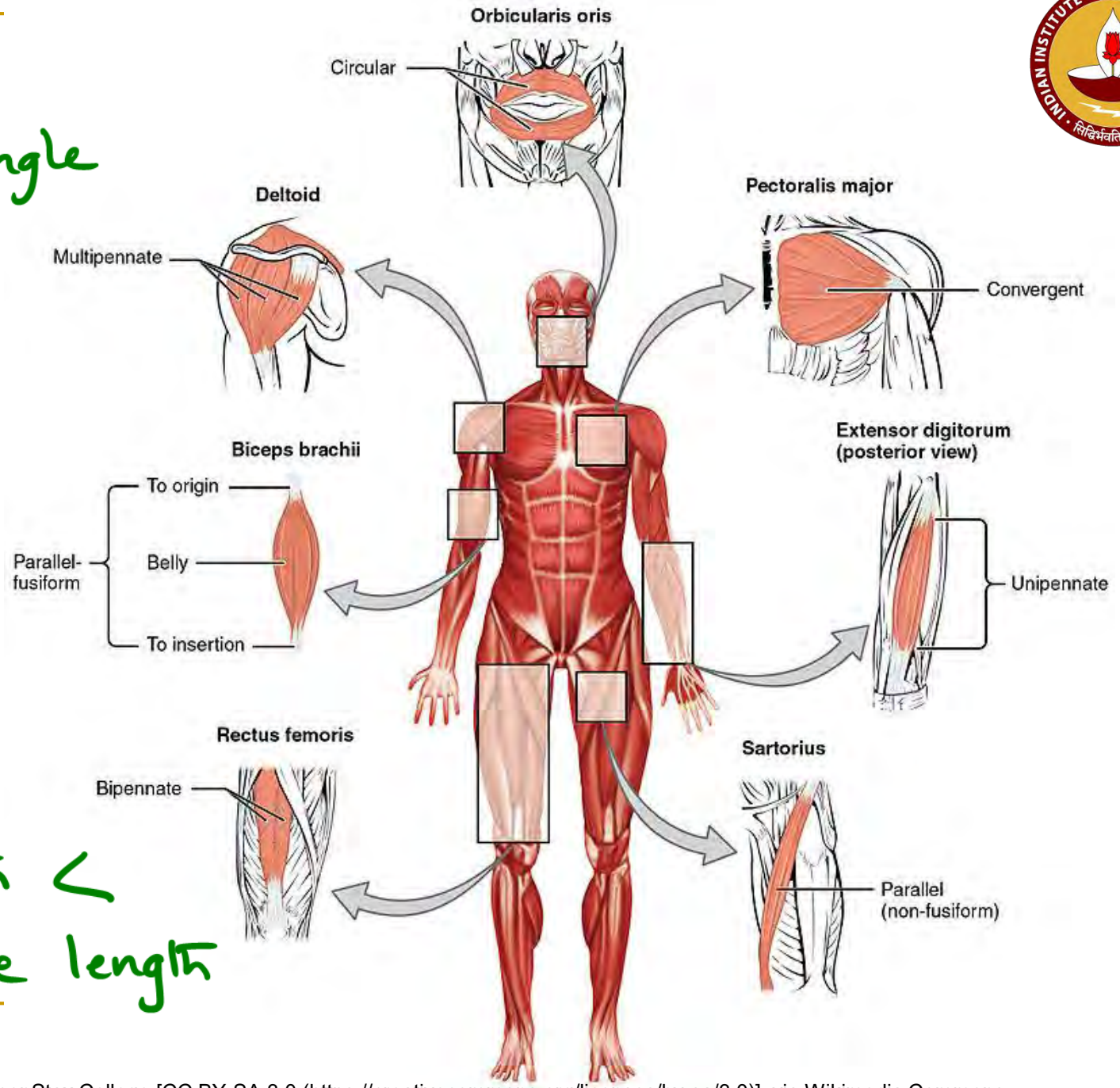
$\theta \rightarrow$ Pennation angle

$F_t = F_m \cos\theta$



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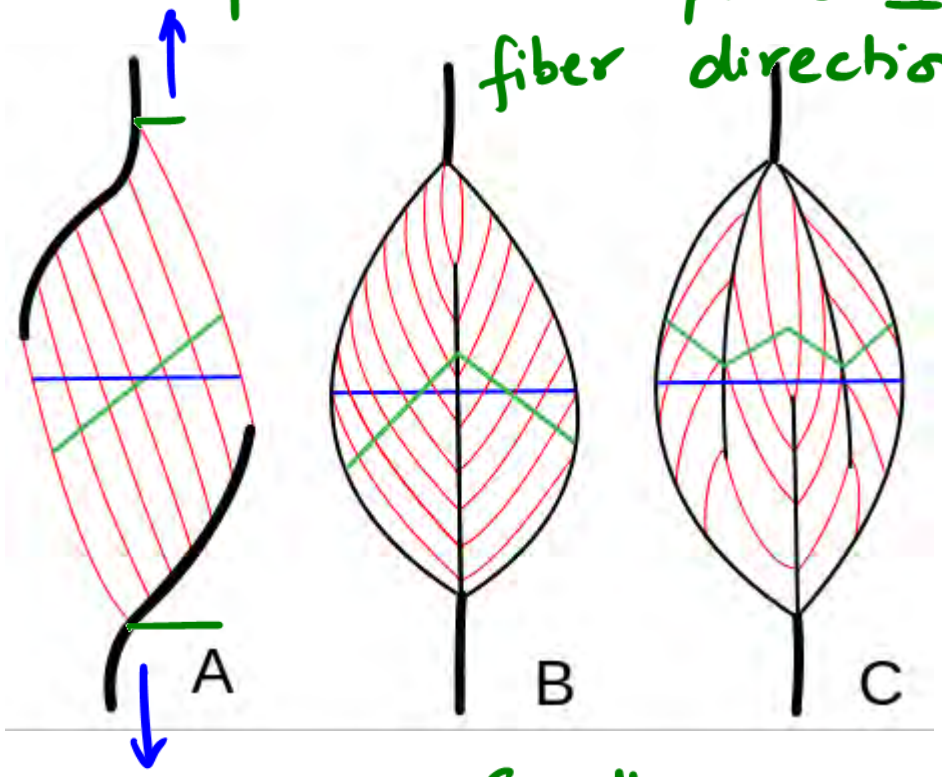
Fiber length < muscle length



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Skeletal muscles

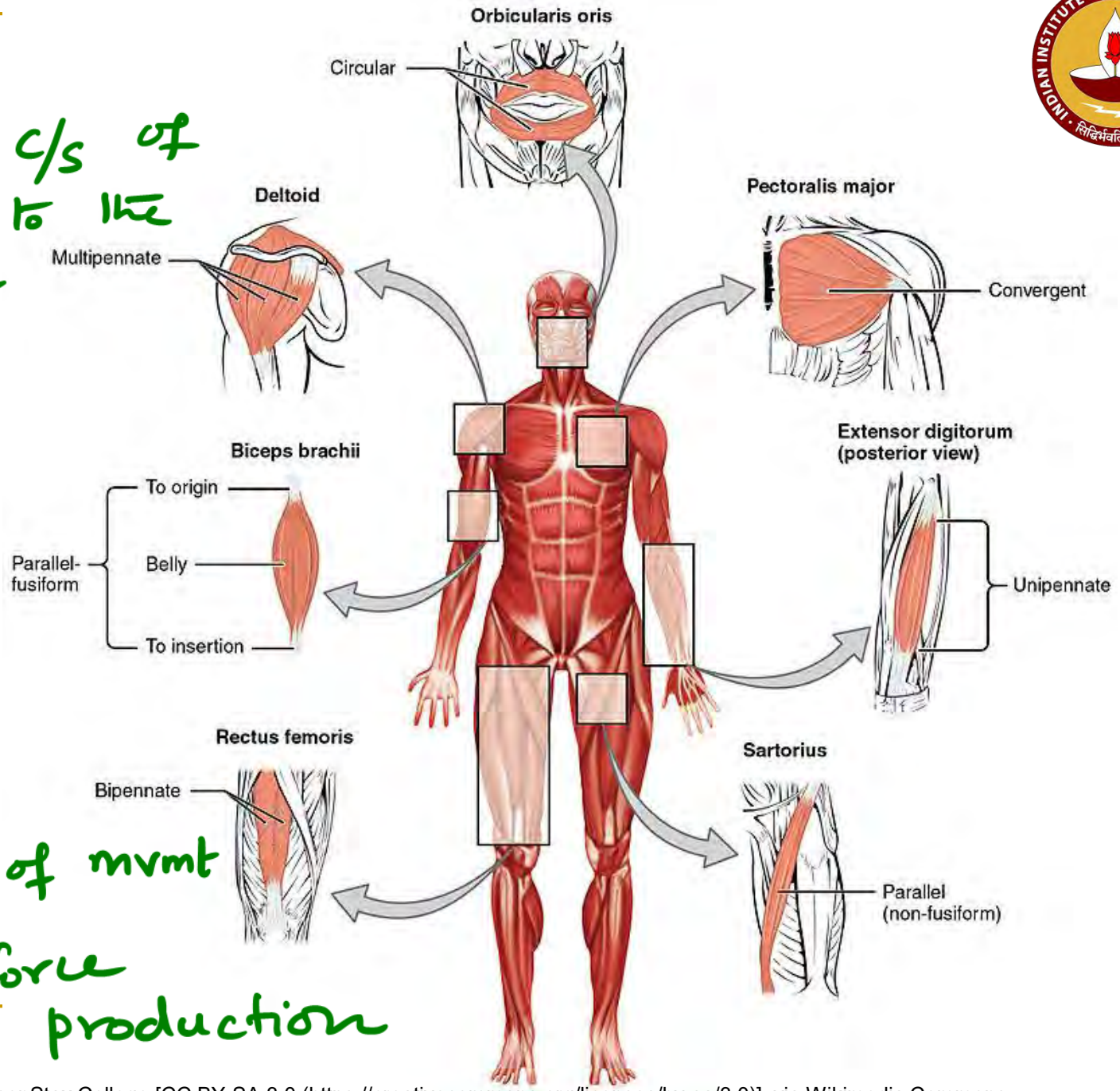
PCSA: sum total of all the c/s of the fibers in the plane ⊥r to the fiber direction



Smaller range of mvmt

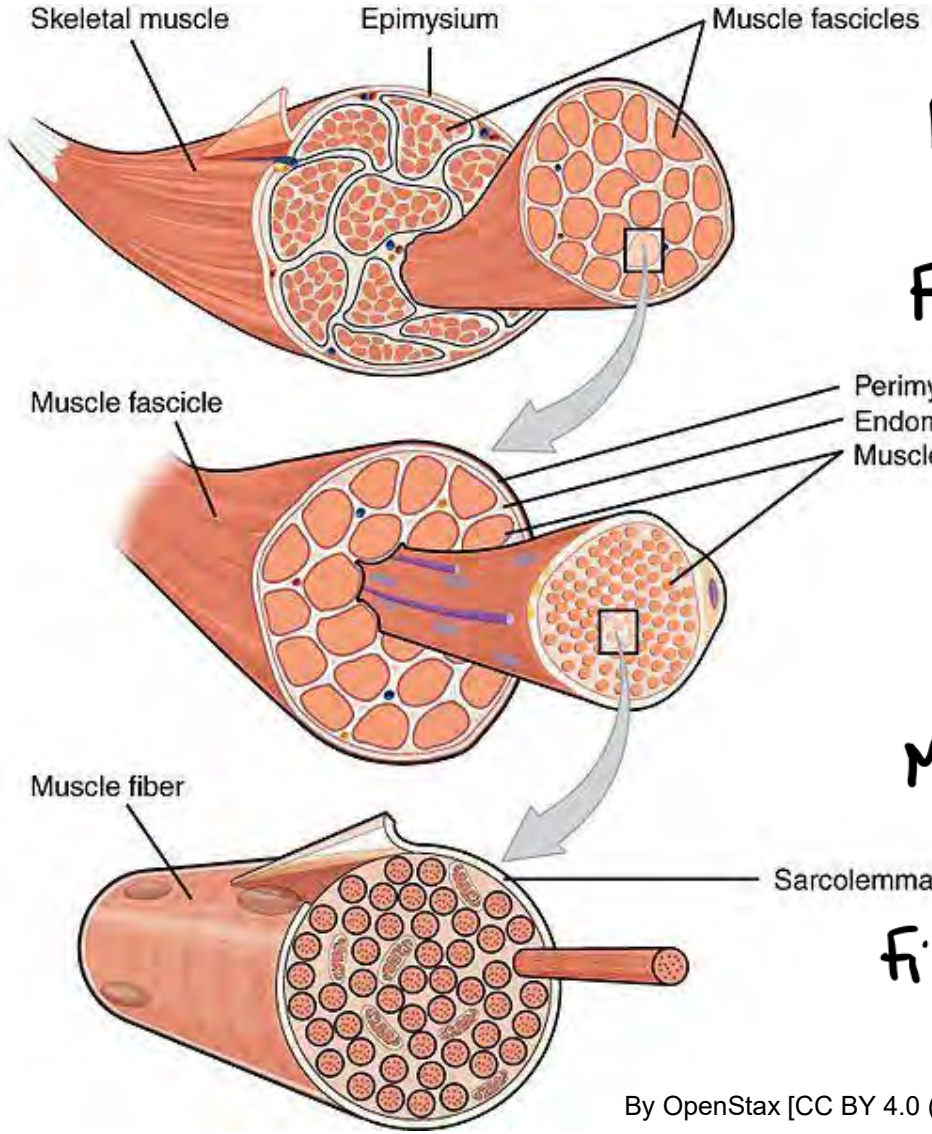
Larger force production

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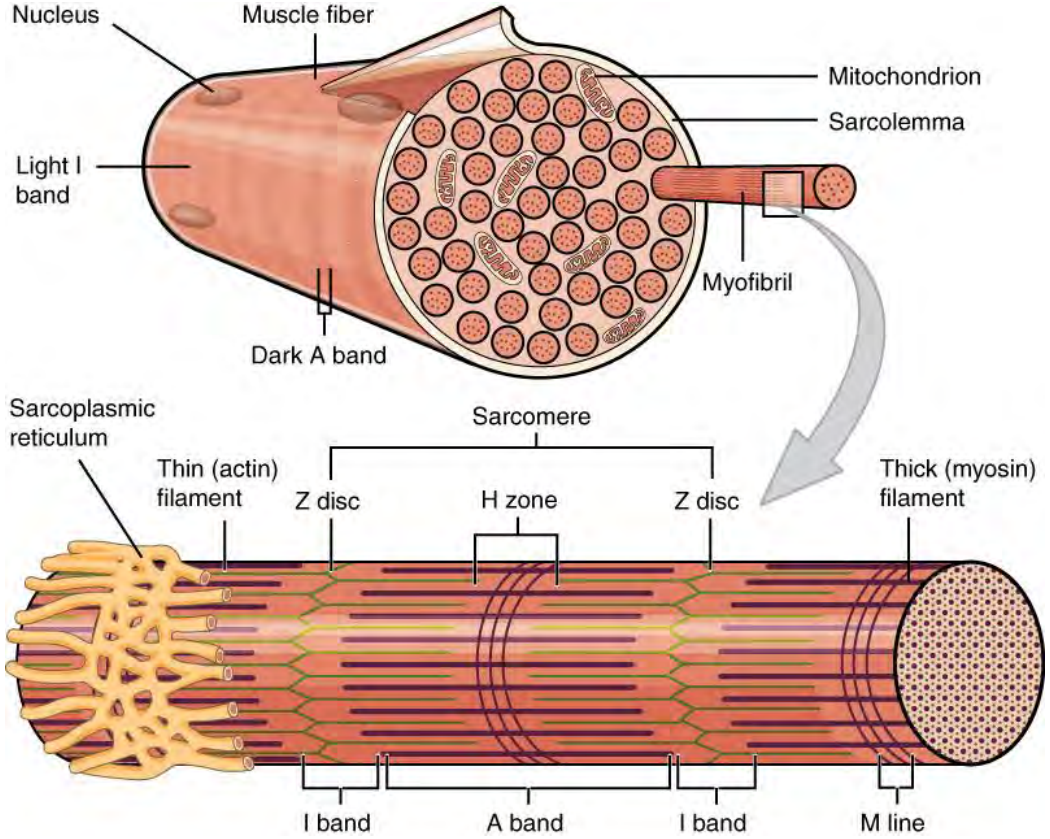


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Muscle fibre structure



Compartments
↓
Muscles
↓
Fascicles
↓
Fibers
↓
Myofibrils
↓
Filaments



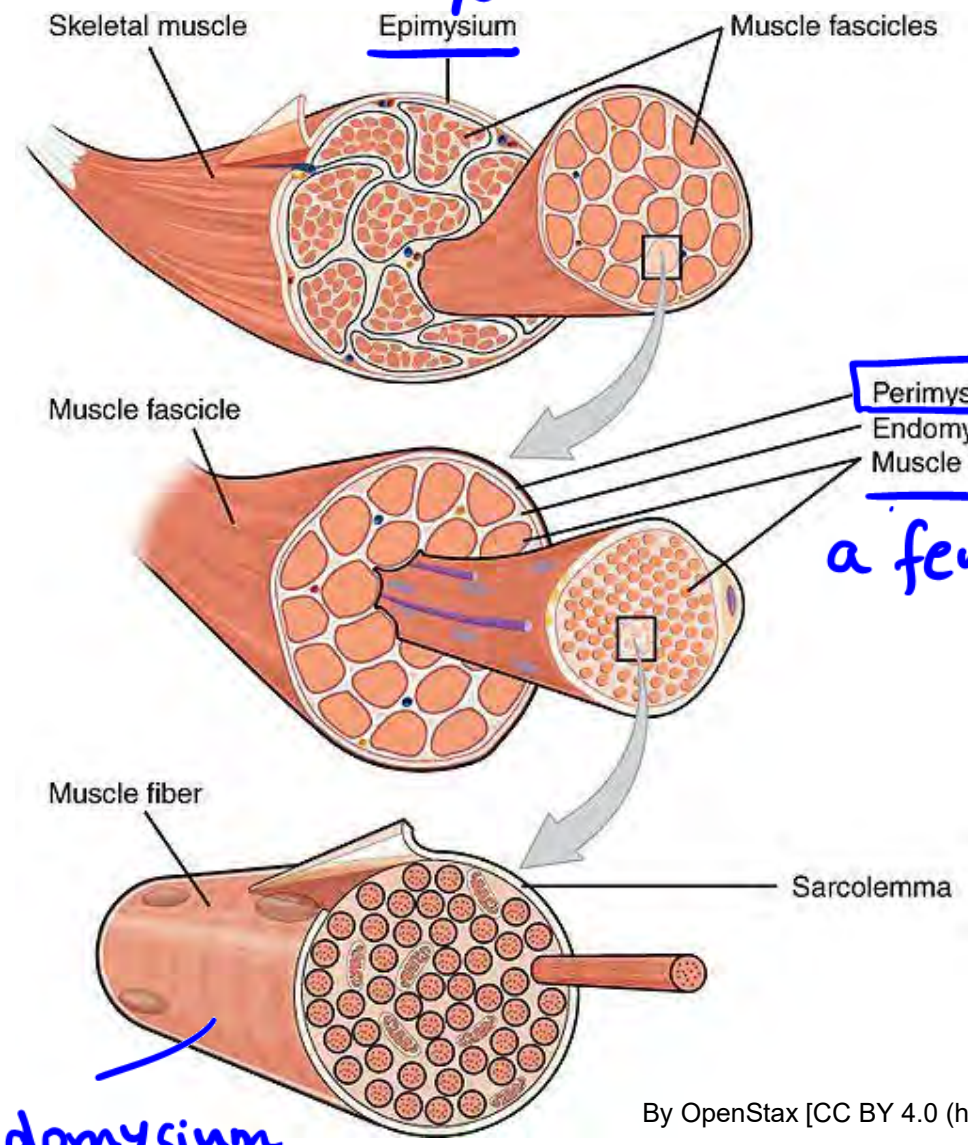
one unit of myosin + actin

SARCOMERE

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Muscle fibre structure

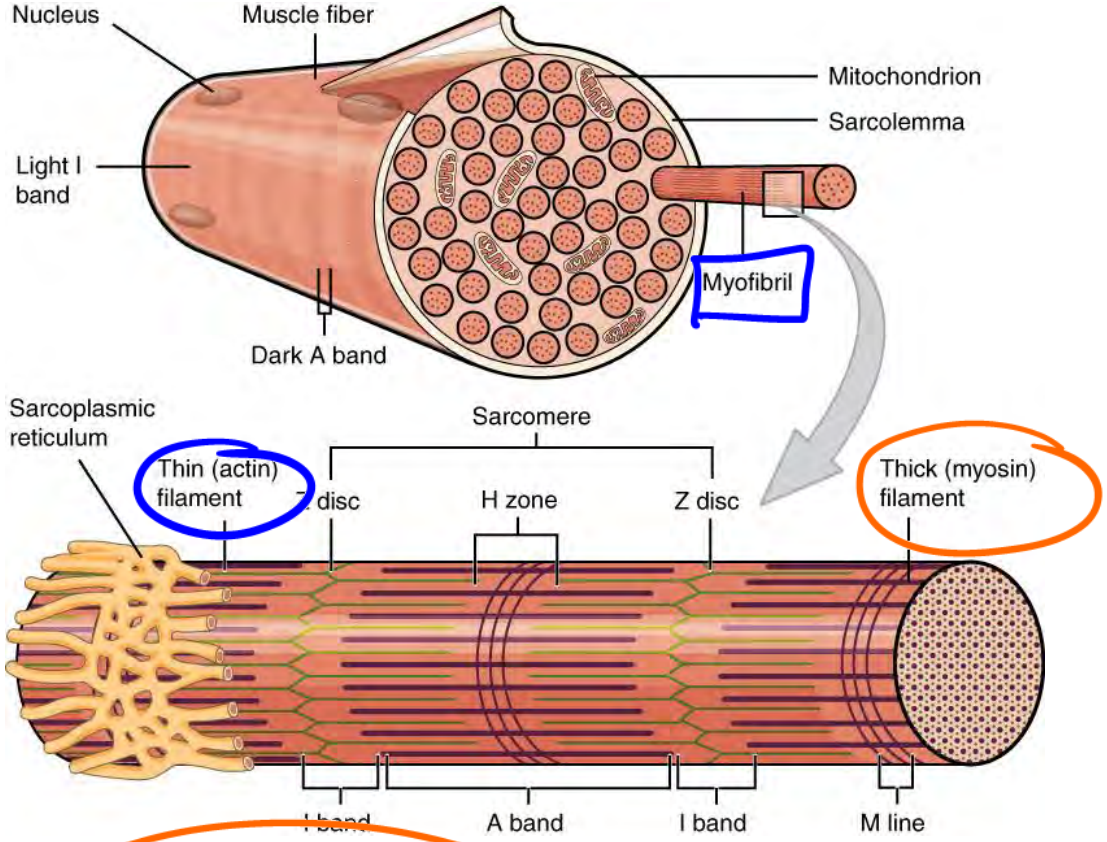
Fascicles are fiber bundles

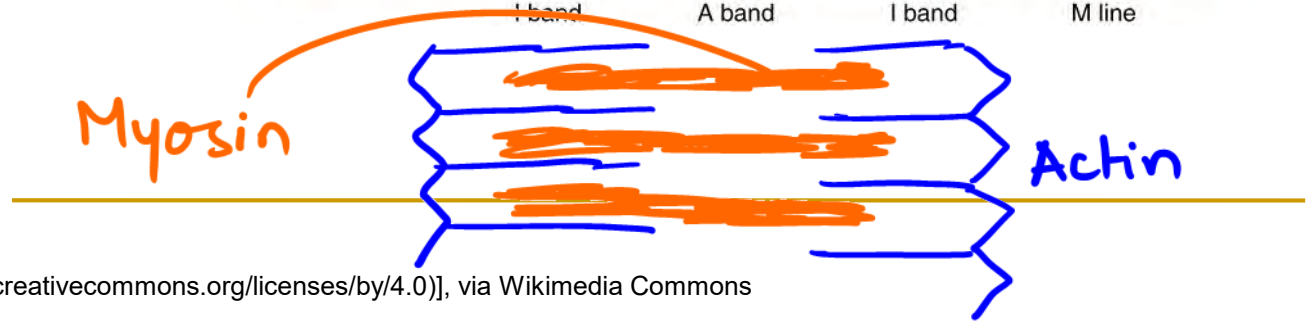


outer fibrous layer

a few hundred

Endomysium

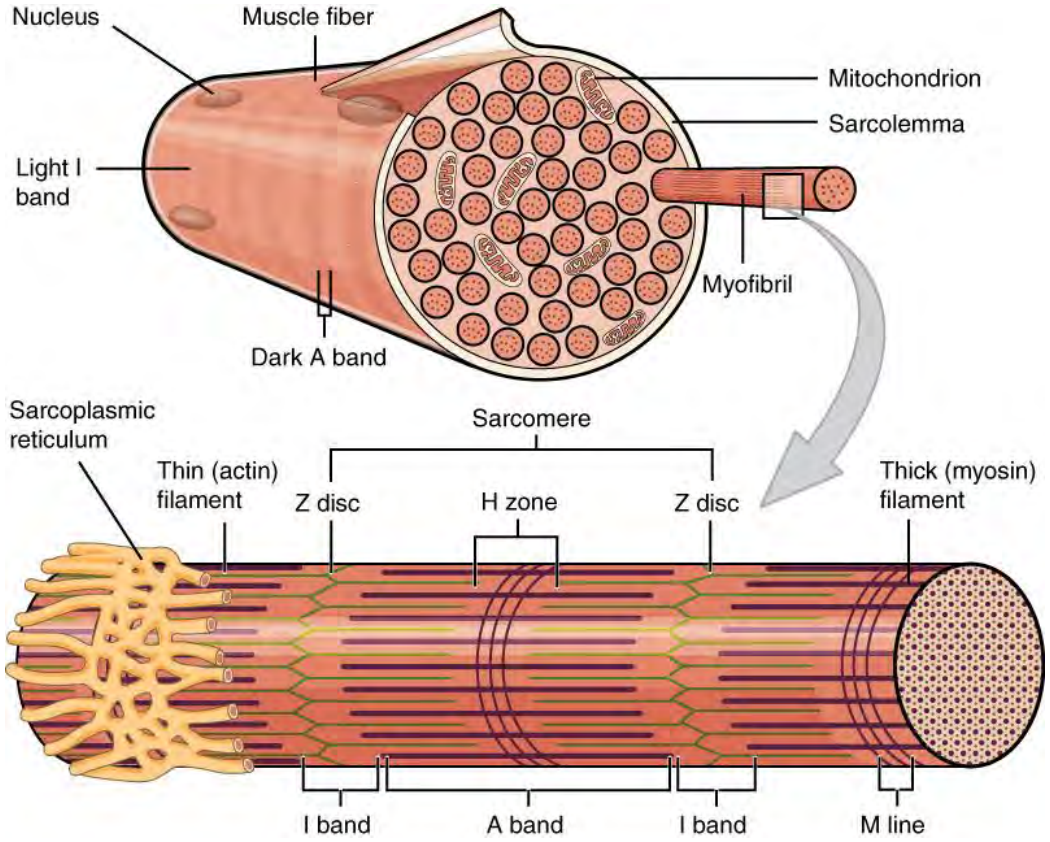


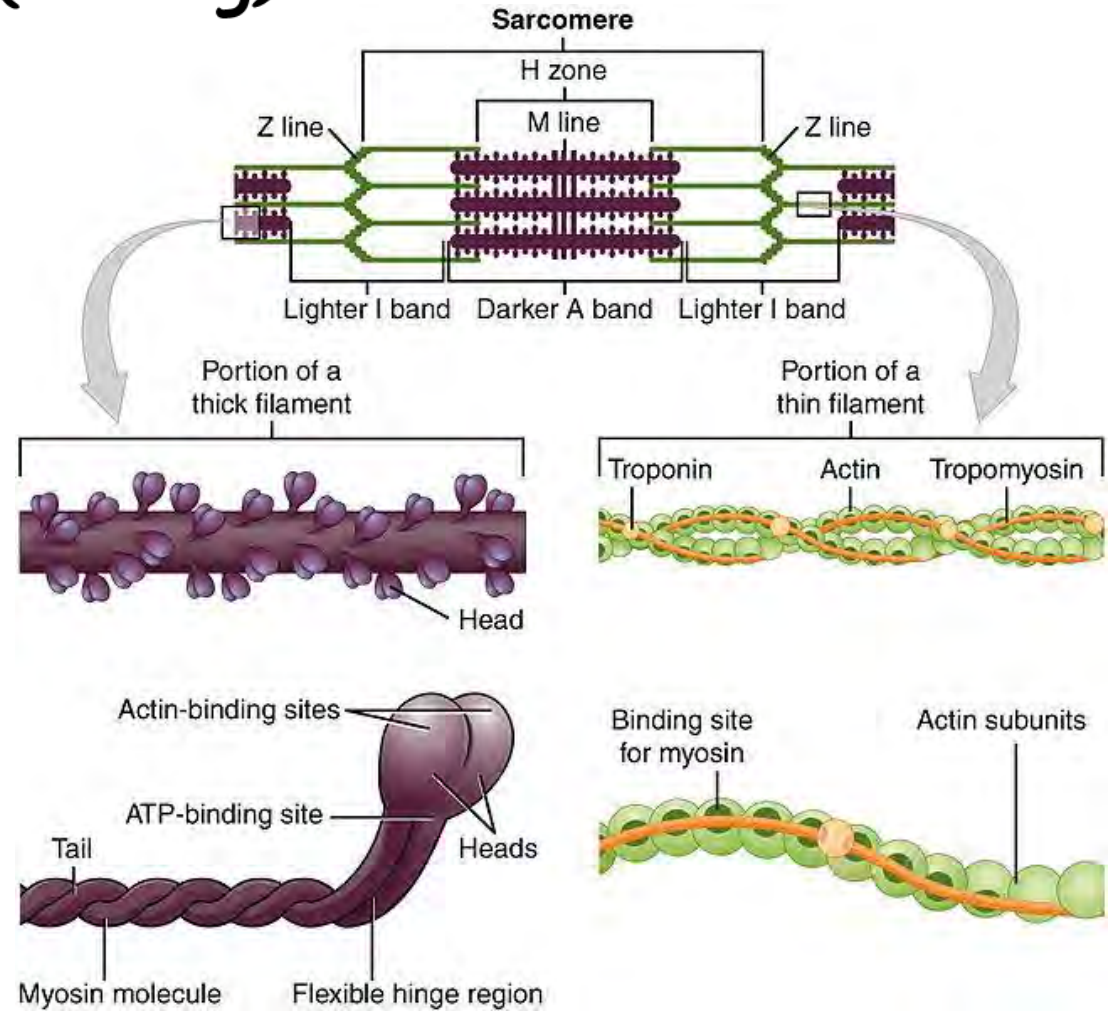


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Muscle fibre structure

Sliding filament theory (Huxley)





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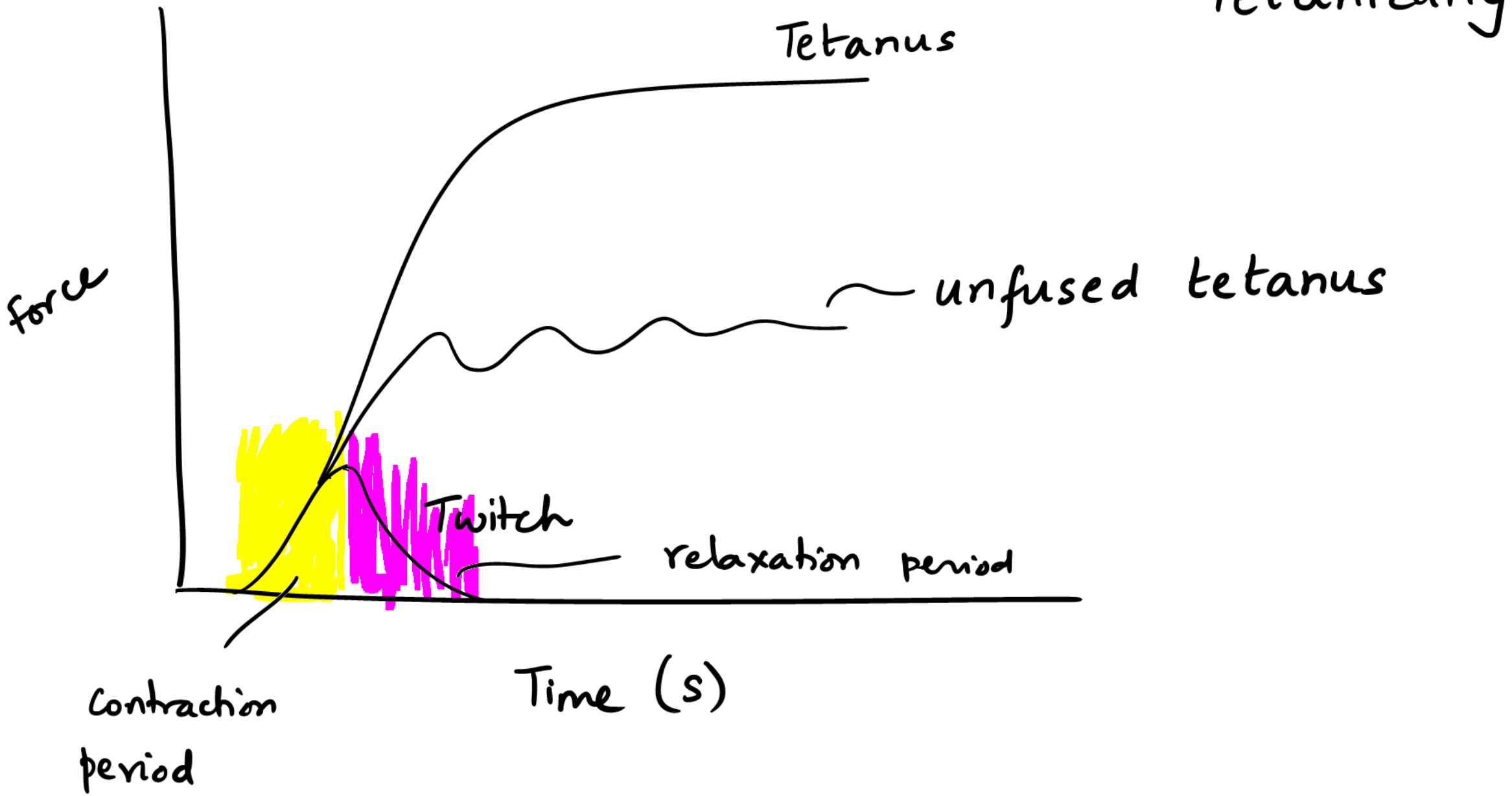
Sliding filament theory (Huxley)

Projections on the myosin form cross-bridges with the actin and during contraction, the thin actin elements slide towards each other

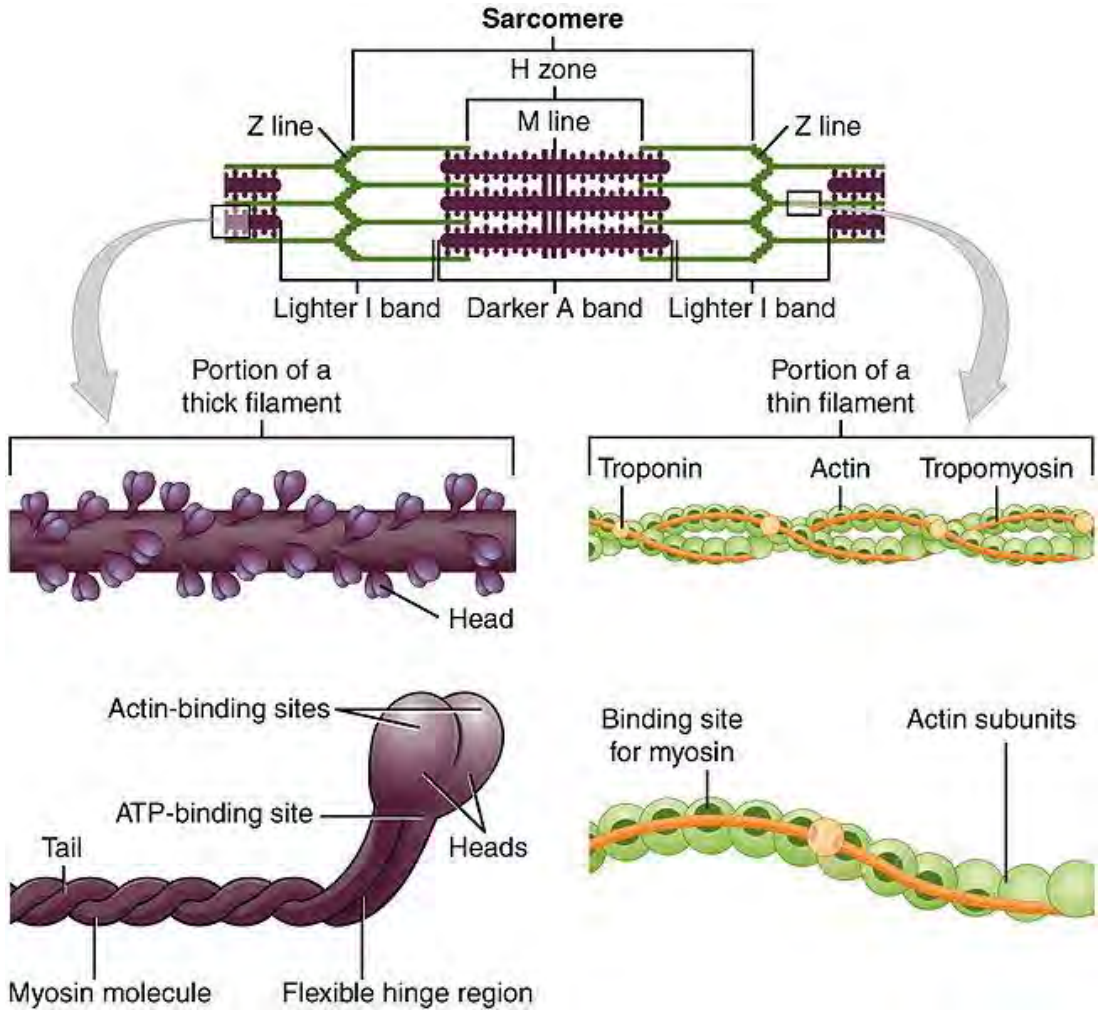
In the contracted state, the actin & myosin filaments overlap along most of their lengths.

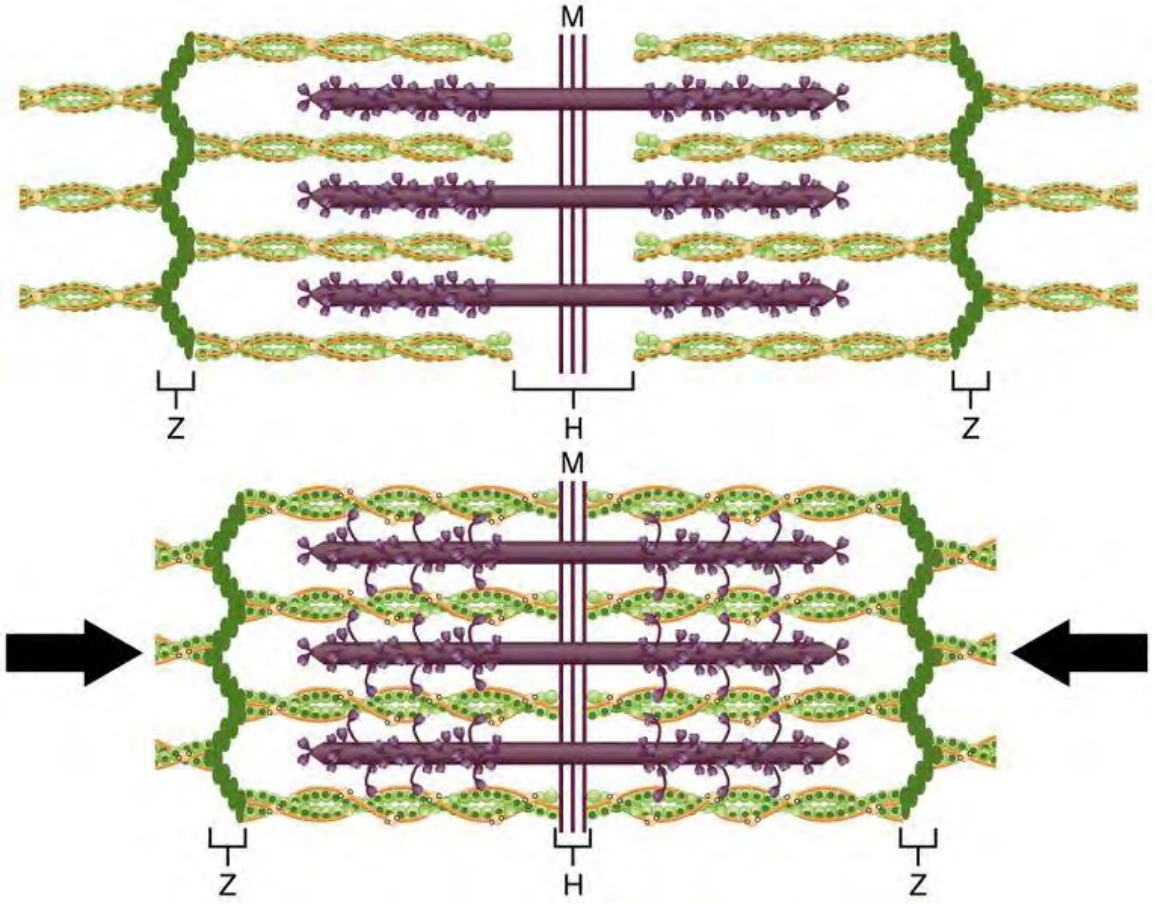
Amount of force developed $\propto$ no. of cross bridges

Tetanically



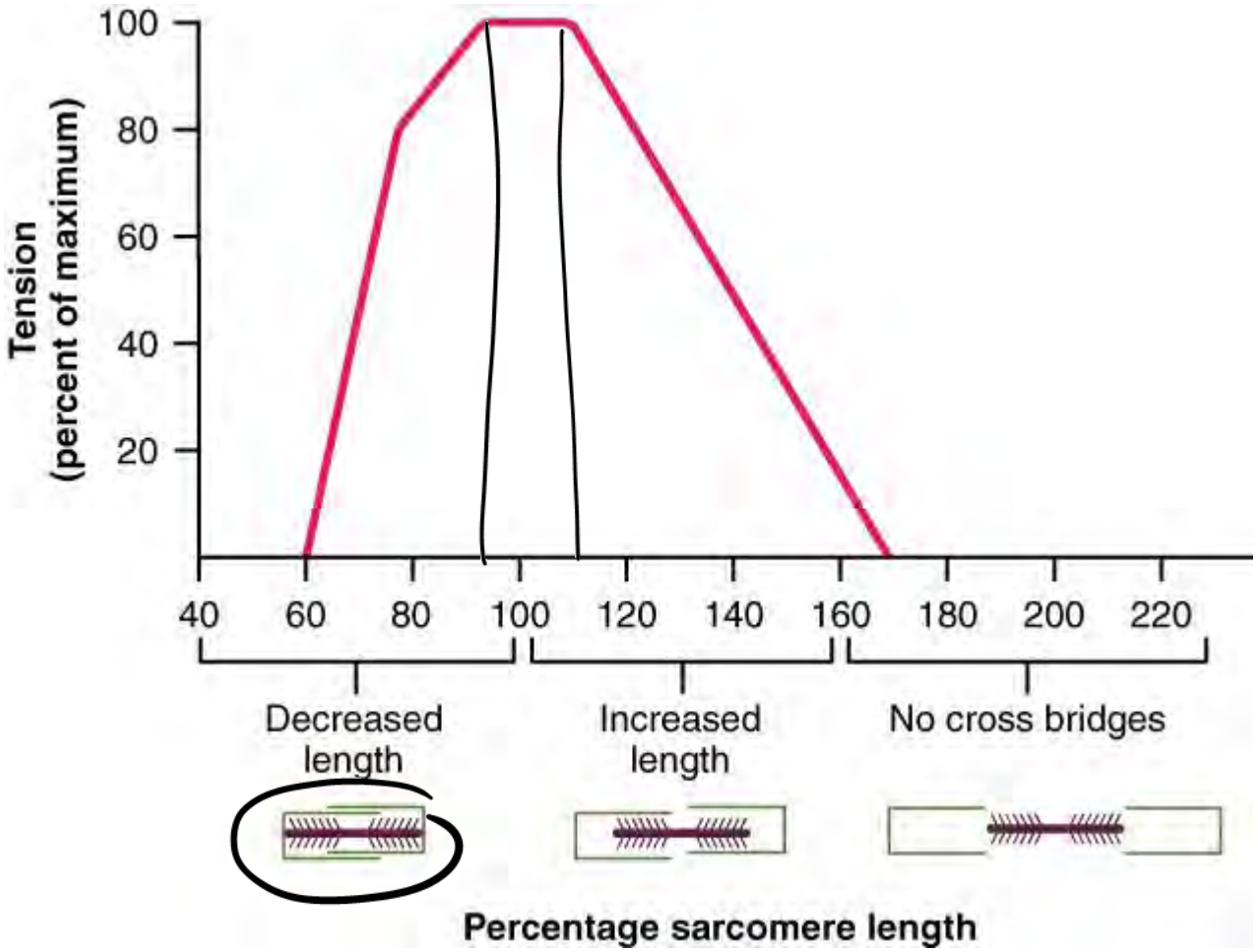
Sliding filament theory





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Length-Tension relationship



Whole muscle behavior will also depend on the behavior of the surrounding connective tissue and the tendons

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